Guest Editorial
Introduction to the Special Section on Open Radio Access Networks: Architecture, Challenges, Opportunities, and Use Cases in Vehicular Networks

CELLULAR vehicle-to-everything (C-V2X) was introduced to support autonomous driving through 5G and beyond networks. C-V2X leverages cellular network infrastructure to integrate vehicle-to-network, vehicle-to-pedestrian, vehicle-to-infrastructure, and vehicle-to-vehicle communications. It has been suggested that Open RAN can be used to achieve the latency requirements essential to realize C-V2X as it achieves real-time optimization through the use of AI in Near real-time RAN Intelligence Controller (Near-RT RIC). The Open RAN will allow the access to historical traffic data or acquisition of data from vehicles. The data will then be transferred to Near-RT RIC for detecting network anomalies while maintaining reliable communication, which is essential for realizing autonomous driving. Open RAN also supports non-real-time RAN intelligent controller (Non-RT RIC) that allows more complex ML workflows such as policy-based feature extraction and optimization to guide vehicles when real-time acquisition is not available. Open RAN provides support for edge cloud, i.e. Open Cloud that helps to interface the Near-RT RIC with Open RAN central unit's user and control plane. Together, the Open RAN and C-V2X are considered to be the key-enabling technologies for achieving low-latency in autonomous vehicular communication networks. The issue attracted over 100 high-quality submissions from all over the world, among which 12 original contributions were eventually selected for publication. The novelty and key contributions of these articles are summarized as follows.

The study by Houran et al. under the title, "Intelligent Reflecting Surfaces Assisted Cellular V2X based Open RAN Communications", proposed a method to improve data transmission in cellular vehicle-to-everything communication using Open RAN and intelligent reflecting surfaces (IRS). The system utilizes clustering and a trellis search algorithm to select the optimal IRS configuration to maximize signal strength between the base station and vehicles. Simulation results show this method outperforms existing techniques in multi-hop IRS communication.

The study by Kumar et al., under the title, "Secure Data Dissemination Scheme for Digital Twin Empowered Vehicular Networks in Open RAN", presnted a new framework called STIoV is proposed to secure communication in IoV networks using Open RANs. STIoV addresses challenges like unverified transactions and data tampering through mutual authentication, reputation scoring, digital twins, and intrusion detection. Evaluations show STIoV's effectiveness compared to existing solutions..

The research proposed a framework for managing UAVs in WSNs for smart agriculture. The framework uses multi-agent deep reinforcement learning to schedule tasks, plan trajectories, and share resources among UAVs. This approach aims to minimize energy consumption and network latency while collecting data from sensor nodes. Simulation results show significant improvement over existing methods, the study by Betalo et al. under the title, "Multi-agent Deep Reinforcement Learning-based Task Scheduling and Resource Sharing for O-RAN-empowered Multi-UAV-assisted Wireless Sensor Networks".

The study by Sun et al. proposed a a new framework called Distilling for Sparse-Meta-transfer Learning (DSML) to address the challenge of limited labeled data in Open RAN-based Cellular Vehicle-to-Everything (CV2x) systems. DSML combines meta-transfer learning with knowledge distillation and sparse-MAML techniques to improve generalization capability and reduce catastrophic forgetting. The proposed method outperforms existing approaches in univariate time series classification tasks, making it suitable for CV2x applications.

The study by Cui et al. under the title, "O-RAN Slicing for Multi-Service Resource Allocation in Vehicular Networks", discussed traditional resource allocation struggles with the growing variety of V2X applications with different needs. This paper proposes a new multi-service strategy for Open RAN-based V2X. It splits resource allocation into two steps: allocating resources between service types and scheduling resources within each type. This allows for specialized scheduling for real-time and non-real-time services, improving both service quality and data transmission rates compared to existing methods.

The study by Cao et al. under the title, "Efficient Resource Allocation of Slicing Services in Softwarized Space-Aerial-Ground Integrated Networks for Seamless and Open Access Services", proposes a framework called Slice-Soft-SAGIN for managing resources in future 6 G networks that integrate space, aerial, and ground networks. Traditional networks offer limited 2D coverage. 6 G with network softwarization (NetSoft) allows flexible resource allocation and service creation.

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Slice-Soft-SAGIN allocates resources across different network types (ground, aerial, satellite) and considers both wireless and wired resources to optimize service delivery for various network slices. The framework's effectiveness is verified through evaluations.

The study by Raja et al. under the title, "UGEN: UAV and GAN-aided Ensemble Network for Post-Disaster Survivor Detection through ORAN", proposed a deep-learning and blockchain enabled secure data processing framework for an edge-enabled green and connected autonomous vehicles. In the proposed scheme, the blockchain ensures the reliability of the vehicles added into the network and the deep learning model is used to detect the intruders in the edge computing environment.

The study by Dalgitsis et al.. under the title, "Cloud-native orchestration framework for network slice federation across administrative domains in 5 G/6 G mobile networks" proposes with advancements in C-V2X and edge computing, network slicing is crucial for guaranteeing performance in connected vehicles. However, maintaining network slices as vehicles move between operators is a challenge. This paper proposes a new framework for network slice federation that allows operators to share resources and maintain consistent slice service for vehicles. The approach is validated through a cloud-native experimental platform showing successful federation and the impact of operator strategies on performance.

The proposed work by Amponis et al. discuss improvements to the QUIC protocol for better data transfer in C-V2X communication using aerial drones. Standard QUIC struggles with frequent network changes in these scenarios. The proposed method adjusts data transfer based on real-time channel conditions, improving performance for C-V2X applications in Open RAN. This paves the way for reliable low-latency communication for autonomous vehicles and future vehicular services.

El Houda et al. proposes a new framework that combines federated learning and deep reinforcement learning to address these limitations. The approach enables distributed training on local data while achieving high accuracy and efficiency in jamming attack detection for Open RAN

Linsalata et al. proposes a new architecture to improve communication between vehicles (V2X) using Open RAN (O-RAN). Traditional RANs lack the flexibility for V2X, but O-RAN offers a solution. The challenge is integrating them seamlessly. This architecture uses a separate low-frequency O-RAN control plane to manage high-frequency V2X communication between autonomous vehicles. Simulations show this approach improves reliability by up to 60% for V2X communication.

The research work presented by Sroka et al. addresses challenges in communication between vehicles (V2X) due to varying service needs and network changes. It proposes Open Radio Access Network (O-RAN) as a solution. O-RAN uses RAN Intelligent Controllers (RICs) to intelligently manage resources for V2X. By leveraging O-RAN's flexibility and special traffic information, the study shows optimized traffic management and resource allocation in V2X scenarios, making O-RAN a good fit for future V2X communication.

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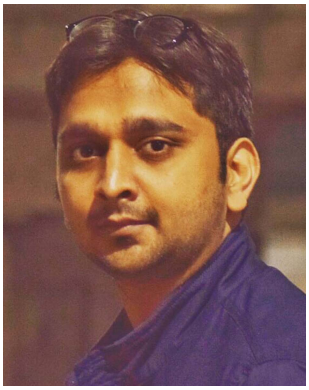

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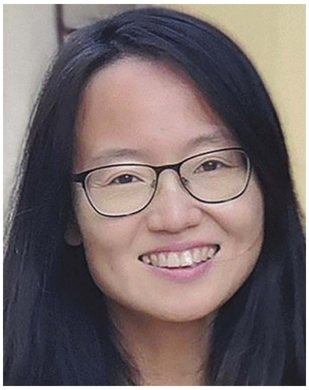

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